THE HONG KONG
UNIVERSITY OF SCIENCE AND
TECHNOLOGY (GUANGZHOU)

Current Verdicts and Legal-Intervention Sufficiency in Battery-Free Design

Development manuscript

20 October 2026

Abstract

Battery-free nodes must acquire energy in time, store usable energy, start above rail thresholds and preserve events until reception. We distinguish a representation that judges a current service from one that supports legal design interventions. Two exposed structural pairs isolate storage saturation and reset retention; saved transient examples connect them to startup, ESR and deadline constraints. Rich static, complete raw and executable-model baselines remain available. A frozen input protocol and discovery-to-consumer contract prepare subsequent Agent experiments. Agent benefit, format benefit, physical qualification and autonomous observation discovery remain separate untested claims.

1 Problem

A fixed BF task T and world w define a service verdict $V(w, d, T)$. A representation $R(w, d)$ is current-verdict sufficient on a declared domain if equality of representations implies equality of current verdicts. For a public legal intervention set $\mathcal{D}$, intervention sufficiency requires the same implication after every jointly legal change $\delta \in \mathcal{D}$ is applied to both designs. The second property need not follow from the first. A passing present trajectory can leave an upper clamp or reset transition unexcited; changing capacitance or power schedule can activate it while the task stays fixed.

The BF question is whether the available description supports the chain from source timing and stored energy through startup, program state and receiving service under a proposed change. The structural witnesses isolate ideal energy-service and event-state guards; the separate transient reference checks receiver events against a deadline. Neither is a calibrated hardware model. This is a question about the information needed by a specific consumer: a static model containing the relevant state laws and parameters can already be intervention sufficient on its declared domain. Acquiring observations is useful only when missing information and its acquisition, qualification and use can be justified.

2 Battery-free mechanism mapping

The two structural pairs isolate BF relations that a passing current trace need not excite: a capacitor's upper energy ceiling and pending-event state across power loss. Their consumers are a timed service-energy transaction and a transmit guard. They are deliberately ideal witnesses, not complete nodes.

A separate saved BF reference supplies startup, transient and event examples (Table 1). The prescribed source first charges a capacitor; ready/off thresholds and boot cost govern startup. A transmit attempt loads the rail according to $V_{\text{load}} = \sqrt{2E/C} - I_{\text{TX}} R_{\text{ESR}}$; a failed rail or energy threshold resets the program. Reset can clear a volatile pending event. This is load–storage–program feedback. The source is prescribed and does

not respond to load, so these examples do not establish source feedback.

Condition	Input (μJ)	First RX (ms)	Event by deadline	Rail gate
Fast pulse	150	15.8	PASS	PASS
Slow pulse	150	54.8	PASS	PASS
High ESR	150	—	FAIL	FAIL
No energy	0	—	FAIL	PASS
Late pulse	150	105.8	FAIL	PASS
Latched reset	210	46.9	PASS	PASS
Volatile reset	210	—	FAIL	PASS

Table 1: Readback of seven saved ideal BF controls; no new simulation. The deadline is 100 ms. A passing rail gate without a receive event does not mean service success. Physical qualification is UNKNOWN for every row.

Fast and slow pulses both supply 150 μJ and pass, with different startup and receive times. The same total energy arriving late misses the service deadline. At 150 Ω ESR, transient droop prevents reception despite the same input energy. With 210 μJ input, a common 40 μJ shunt and postponed TX, the latched and volatile cases differ in event survival. The passive mechanical latch explicitly captures at zero electrical energy and retains across reset; this assumes neither free powered NVM writes nor a calibrated commercial latch. All sources and devices remain uncalibrated. Agreement with a predeclared negative control is distinct from a passing event-service verdict.

3 Observation acquisition

Observations add facts at measured acquisition and qualification cost. We distinguish selection of developer-defined observations from constructing executable observations. A static sufficient statistic or a complete executable model may provide all necessary information.

4 Implementation

The frozen protocol specifies ideal clamped energy and reset-state models, legal D changes, separate reference implementations and negative controls. Phase one reads and audits the saved results, adds BF mechanism mappings and participant input gates, and freezes a later Agent discovery contract. It changes no existing physical or task semantics and reruns no historical solve.

5 E2–E5: public development results

5.1 E2–E3: two unexcited mechanisms

Run `IF-7412532c4a6ab7a5879e` contains two public development pairs, not statistical replications or unseen tests. Each pair has identical current observed event/rail traces and two passing current tasks. The identical legal D change then yields FAIL/PASS.

For the clamp pair, a fixed 20 μW source supplies 40 μJ before a 30 μJ service transaction at 2.5 s, with a 1 V service reserve. Both systems use 10 μF initially. Their ideal clamps are 3 V and 5 V. The initial maximum rail voltage is $\sqrt{8} = 2.828$ V, so neither clamp activates and the entire current trajectory coincides. Changing only capacitance to 4 μF is legal in the common $[4, 12]$ μF domain. The 3 V clamp now spills 22 μJ and makes only 16 μJ available above reserve; it fails the fixed service. The 5 V clamp spills zero and delivers 30 μJ; it passes. The post-change margins are -14 and 8 μJ. These are ideal energy transactions, not measured PMIC startup or MCU current.

For the reset pair, an event is captured at 1 s and must be transmitted once at 4 s, by a 5 s deadline. The external event and transmission time stay fixed. Without a power reset, volatile and retained pending

memories both deliver the event. A common D change inserts an outage from 2 to 3 s: volatile pending memory is cleared and delivers zero events; retained pending memory delivers one. This changes a state transition rather than rescaling an old energy/window template.

The analytic clamp implementation agrees with projected time integration; the ordered memory machine agrees with a separate event-program parser and tick interpreter. Both 1 ms and 0.5 ms settings agree on all eight current/intervention cases. Maximum energy discrepancy is 2.84×10^{-12} μJ. These implementations share ideal assumptions, so agreement is numerical/software evidence only.

Deleting the unexcited clamp threshold or reset-retention information restores aliasing in this encoding. It does not establish universal necessity of a field name or complete device-level E3 validation. Indeed an enriched *static* model is sufficient on the declared ideal domain. It passes all 28 exposed legal-grid checks (20 clamp, eight memory), and a purely additive static control agrees with independent packet accounting on five service-cost values, including passing and failing designs. The clamp statistic is

$$M(C) = \min(E_0 + P_{\text{in}} t_{\text{in}}, CV_{\text{max}}^2/2) - CV_{\text{min}}^2/2 - E_{\text{service}}.$$

Its sufficiency follows here from all input preceding service, an ideal cap, and no leakage or intervening workload. Those assumptions bound the claim.

5.2 E4: executable equal-information preparation

At each information level the prose and JSON packets contain identical facts and an identical summary (eight parity checks across 16 packet files). In fixed-information use no additional tool or read is legal. In active-observation use all arms start with the same current packet and share the same legal observations. Table 2 reports deterministic parse-and-judge checks on the proposed intervention. Abstentions are not counted as errors. This constrained prose serialization is preparation, not a test of model comprehension or unconstrained writing.

Information	Format	Correct	Unknown
current	free-text	0/4	4/4
current	structured	0/4	4/4
sufficient	free-text	4/4	0/4
sufficient	structured	4/4	0/4

Table 2: Fixed-information parser/decision checks; no model calls. Model-format benefit is NOT_RUN.

5.3 E5-1: measured finite observation selection

The selector receives researcher-supplied required-field metadata, not outcomes, and chooses the smallest eligible source by actual file-size estimate. Table 3 totals four public cases. Bytes are payload bytes returned to Python, separately counting source readback for audit; they are not storage-device I/O or monetary fees. Wall time includes acquisition, source audit, and selector where used. Source-export setup is retained in the raw data. These are single host measurements with uncontrolled OS cache, not a latency benchmark.

Observation	Correct	Read B	Audit B	Wall ms	Model runs
uninformative	0/4	4648	4648	0.289	0
mechanism-spec	4/4	103	103	0.260	0
fixed-rich	4/4	4747	4747	0.282	0
full-raw	4/4	8544	8544	0.333	0
executable-dynamic	4/4	8544	8544	1.886	4
selected	4/4	103	103	0.236	0

Table 3: Observation acquisition and audit costs. Uninformative returns UNKNOWN on all four.

Fixed-rich, complete raw information, and an executable dynamic response all judge four of four correctly. The selected observation reads exactly the same facts and bytes as the fixed minimal mechanism-spec baseline. This batch supplies no independent algorithmic selection gain and does not weaken those strong baselines. A zero-byte acquisition budget abstains; three illegal changes and a tampered source binding are rejected.

5.4 E5-2: executable construction preparation

A deterministic compiler turns a researcher-specified headroom expression into a new standalone Python observation, with source hash, input schema and applicability domain. Its values -14 and 8 μJ agree with independent integration on all 20 exposed clamp-grid points and in two fresh-process executions. Construction took 0.239 ms; reference qualification took 13.017 ms (20 model runs); the two fresh processes took 37.730 ms combined. The equation and relevant inputs were manually supplied. This validates executable construction and audit machinery, not autonomous discovery or held-out usefulness.

Evidence binding. The frozen protocol, raw result, packet files, generated program and costs accompany `IF-7412532c4a6ab7a5879e`. Data source: `docs/research/parallel-20261019/interfaces/data/IF-7412532c4a6ab7a5879e/result.json`. Entry point: `scripts/campaign_interfaces.py`; manuscript tables: `papers/interface-sufficiency-20261019/render_results.py`. Paid model requests and native dispatches are zero. Historical successful product solves were not rerun. The old three scaled families and the refinement/fixed-rich tie remain separate development negatives.

6 Frozen E4 input and resource protocol

Phase one exports **64 participant inputs**: four existing cases, four information levels, two formats and two observation regimes. All **32 format pairs** have identical decoded facts, summary, assets, tool definitions and resource limits. The main factorial uses the current interface and a BF-rich static model. Complete raw data and executable response are additional strong baselines. This is an input-integrity result. E4 live Agent-format trials are NOT_RUN.

The static model includes actual-world parameters, state rules, service windows, units, assumptions and legal design domain. It is sufficient for the declared ideal witnesses: the 28 static-grid and five additive controls are retained saved results. Complete raw preserves every physical field, law, current output, alternative world and public grid; only historical identifiers are removed. The executable baseline includes a standalone export of the unchanged numerical model bodies. Preparation checks its syntax without running it. No baseline is reduced to total energy or average power alone.

Each trial has one participant directory. Future-intervention labels and historical case mappings live in a separate evaluator index. The participant input gate reads only that trial directory, verifies packet/asset hashes and rejects unlisted files, private paths and explicit answer/relevance metadata. Current verdicts and physically informative parameters remain legal facts. These public examples were already exposed; the directory split does not make them unseen tests or provide an operating-system sandbox.

In fixed-information use, external observations are prohibited and local computation on supplied assets is allowed. In active-observation use, the same initial packet at each information level is combined with a common API for current trace, full declarative specification, complete raw model and model response. Newly acquired facts, source digest, units, bytes and wall time are recorded separately. This phase replays saved responses only; a different design query abstains rather than launching a new solve. Live trials require fresh contexts, one-directory mounts, an externally owned cumulative ledger, and identical per-trial limits. The registered profile disables model, Agent and native dispatch; no paid/provider lease is admitted here.

7 E5: Agent discovery to a real BF consumer

E5-1 asks an Agent to choose a legal observation from a general catalogue. E5-2 asks it to identify a missing BF relation and propose the observation and executable implementation itself. The initial interface supplies the task, current facts, legal interventions and general tool schemas/costs. Relevant-field labels, target

observations and future service labels remain evaluator-only. The earlier required-field selector and human headroom expression are explicit construction baselines, not discoveries.

The frozen contract requires six stages: (1) a provenance-bound Agent proposal; (2) a legal acquisition with units, domain and budget; (3) an immutable observation bound to source, code, full input and actual cost receipts; (4) qualification by a separate implementation, with static-rich, uninformative, tampered and out-of-domain controls; (5) a typed input edge into a kernel-registered BF consumer, bound to its full input, code, execution and independent service check; and (6) an outcome-blind next judgment or design decision that retains rejection or UNKNOWN when appropriate. Provider, model, run and usage identity must come from external authority. Self-declared Agent ownership, self-reported costs, echoing an observation or signing one's own consumer receipt is insufficient.

The existing loaded-rail/reset/event program is a candidate consumer; an adapter connecting a qualified relation to that real program is not registered in this phase. The validator therefore admits no real E5 experiment without external kernel authority. Synthetic registry unit fixtures test admission rules only. Even a valid receipt can record service FAIL; it establishes no improvement. A benefit claim needs a separate frozen with/without-observation comparison. Genuine Agent discovery, new independent qualification and real-consumer use are all NOT_RUN; transfer and hardware remain untested.

Phase-one binding. The preparation bundle is `BFP-1516e936ff2a3bd8902e`. Its data and contract are stored under `docs/research/phase1-20261020/interfaces/data/`. Preparation and gates: `scripts/campaign_bf_interfaces.py`. New solver runs, paid model requests and native dispatches are zero. The two original structural witnesses, 16 packets, numerical checks and historical costs remain unchanged.

8 Boundaries and threats

All examples are exposed public development mechanisms. Two implementations of the same ideal equations check implementation consistency, not device reality. Fixed-rich, full raw information, executable dynamic models, and sufficient static controls remain valid baselines. No novelty priority is claimed without a literature audit. Historical three-family refinement tied the fixed-rich baseline and remains a separate development result.